Cardwell ZR-75-AS Trim-Air Variable Capacitor

The Allen D. Cardwell Manufacturing Corporation / Cardwell Condenser Corp

Product Overview

The Cardwell ZR-75-AS is a precision Trim-Air series variable capacitor engineered for high-frequency VHF/UHF oscillator, receiver, and transmitter trimmer applications. Built with a high-grade steatite end plate, it minimizes dielectric absorption and dielectric heating while maintaining mechanical stability under vibration.

Technical Specifications

Model Number	ZR-75-AS
Series	Cardwell Trim-Air (ZR Series)
Maximum Capacitance	75 pF (mmf)
Minimum Capacitance	3.0 pF (mmf) nominal
Nominal Air Gap	0.030 inches
Voltage Breakdown Rating	1,000 Volts Peak (Test Rating)
Insulation Material	High-frequency glazed Steatite / Isolantite ceramic end plate
Plates Material	Precision stamped aluminum with rounded polished edges

Key Features & Construction

- Isolantite / Steatite ceramic end plate ensures low RF loss at VHF/UHF frequencies
- Slotted rotor shaft permits screwdriver alignment adjustment with panel locknut
- Integral threaded bushing allows single-hole panel or chassis mounting
- Phosphor bronze wiper contact ensures consistent low-noise rotor grounding
- High electrical Q factor suitable for exciter, VFO, and tank circuits

Mechanical & Mounting Details

- Mounting: Single-hole panel mounting via 1/4-32 or 5/16-32 threaded bushing and hex locknut
- Rotor Shaft: 1/4 inch diameter shaft with slotted end for screwdriver adjustment
- Plate count: Multi-plate semicircular straight-line-capacity rotor/stator assembly
- Base dimensions: approx. 1.25 x 1.50 inches ceramic frame